

Matthew Vaishnav

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SUMMARY

Computer Systems Technician student (Conestoga College) seeking co-op in SOC analysis, DevSecOps, or systems administration. Hands-on experience running an 18-node home lab across 6 isolated VLANs with pfSense routing, Security Onion IDS, and full IaC via Ansible and Terraform. Projects include a production anti-DDoS system validated at 96% accuracy on the CIC-DDoS2019 dataset, a DevSecOps CI/CD pipeline with SAST/container scanning, and CTF writeups with paired Sigma detection rules mapped to MITRE ATT&CK.

Portfolio: matthewvaishnav.github.io/portfolio

PROJECTS

SENTINEL — Anti-DDoS System

2025–2026

github.com/matthewvaishnav/sentinel

- Production-grade Node.js/Express anti-DDoS engine validated at 96% accuracy, 98.33% recall on the CIC-DDoS2019 industry dataset
- Async math worker pool offloads all $O(N^2)$ neural net backprop and FFT analysis off the event loop, keeping HTTP response times stable under volumetric attack
- Dynamic Z-score + EMA self-learning baselines replace static thresholds; 96.41% precision calibrated to minimize false positives on legitimate traffic
- WebSocket P2P gossip mesh propagates threat blocks across regional instances in milliseconds with proof-of-threat consensus and no central authority
- Horizontal Redis clustering for cross-region state; L1/L2 cache for instant IP reputation lookups; CI/CD pipeline with GitHub Actions cross-platform binary builds

Computational Pathology Research Framework

2026

github.com/matthewvaishnav/computational-pathology-research

- PyTorch ML framework for whole-slide image (WSI) analysis; implements PatchCamelyon training pipeline and CAMELYON16-style slide-level classification
- OpenSlide integration reads .svs/.tiff/.ndpi formats; patch extraction with tissue thresholding; ONNX model export for deployment
- CI/CD with Bandit SAST, isort, pre-commit hooks; 62% test coverage; Docker + docker-compose; pyproject.toml packaging
- Supports torchvision and timm encoders (ResNet, EfficientNet, ViT); mean/max pooling MIL aggregation; CSV prediction export with MITRE-style reporting

Drift — Server State Tracker

2026

github.com/matthewvaishnav/drift

- Git-like server state tracker for tracking configuration changes, rollback capabilities, and known-good state management
- Python CLI tool for Linux/macOS/Windows; tracks what changed, when it changed, and how to restore previous configurations

TECHNICAL SKILLS

Security	Security Onion, Suricata, Zeek, Sigma, Splunk SPL, MITRE ATT&CK, Elastic SIEM, Azure Sentinel, pfSense, Wireshark
DevSecOps / Cloud	Terraform, Ansible, Docker, GitHub Actions CI/CD, AWS (VPC, GuardDuty, CloudWatch, S3), Kubernetes (k8s), Bandit SAST, Trivy
Systems	Linux (Ubuntu, Debian), Windows Server 2019, Active Directory, VMware Workstation, VLAN design, pfSense, TCP/IP, DNS, DHCP
Programming	Python, Bash, PowerShell, JavaScript (Node.js/Express), C++, PyTorch, SQL
Tools	Git, Redis, Prometheus, Grafana, Loki, OpenSlide, ONNX, Docker Compose, Raylib, Adobe Premiere/After Effects

EDUCATION

Computer Systems Technician — IT Infrastructure & Services

Sept 2025 – Present

Conestoga College | Waterloo, ON

- Hands-on curriculum: hardware diagnostics, Linux/Windows administration, firmware, Docker, network infrastructure
- Working toward co-op eligibility — available Summer / Fall 2026

High School Diploma

Sept 2020 – June 2025

Grand River Collegiate Institute | Kitchener, ON

CERTIFICATIONS & CREDENTIALS

- **CompTIA Security+** — In progress — 62% complete
- **TryHackMe SOC Level 2** — In progress — 78% complete
- **TryHackMe Pre-Security** — Completed
- **Cisco Networking Essentials** — Completed
- **Royal Conservatory of Music — Piano Level 3** — Prep A & B with Honors and Distinction; 8+ years practice
- **First Aid Certification** — March 2024 – Present
- **WHMIS 2015** — Certified
- **Class G Driver's Licence** — March 2025 – Feb 2028

WORK EXPERIENCE

Marketing Co-op Student

Apr – June 2024

United Way Waterloo Region | Kitchener, ON

- Assisted planning and execution of fundraising events; conducted donor market research; contributed to social media campaigns

Adventure Specialist

Aug 2021 – Aug 2023

Adventure Rooms Canada Inc. | Kitchener, ON

- Managed game room setup/reset between groups, handled payments and reservations, and created social content for X/Twitter and Facebook
- Repaired and replaced props and game components; performed hands-on facility maintenance including hardware repairs and paint touch-ups